THARUN G

Space Systems Security | AI | Earth Observation

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SUMMARY

Third-year CSE student focused on securing and monitoring space systems. Built an explainable anomaly-detection system for satellite telemetry (Smart India Hackathon 2026 entry) and an autonomous SIEM/SOAR assistant, works with Landsat and Sentinel imagery, and plays web-security CTFs. Seeking space-tech, SOC or security engineering internships.

EDUCATION

B.E. Computer Science and Engineering | National Engineering College, Kovilpatti 2024 – 2028 (Expected)
CGPA: 6.8 / 10

SKILLS

Space & Earth Observation: Satellite Telemetry (SatNOGS), Anomaly Detection, Google Earth Engine, Landsat, Sentinel-1/2, QGIS, GDAL
Security: Web Application Security, Vulnerability Assessment, Threat Modelling, Burp Suite, Nmap, MITRE ATT&CK, SIEM/SOAR, Sigma, OCSF
AI & Engineering: Python, PyTorch, scikit-learn, LangGraph, Ollama, Docker, Linux, Git

PROJECTS

Astrovia: Explainable Satellite Anomaly Detection | Python, PyTorch, scikit-learn, SatNOGS, Streamlit, Plotly, Docker SIH 2026 (Waitlisted)

- Replaced fixed-threshold telemetry alarms with per-channel PyTorch LSTM forecasters and z-score scoring of prediction residuals; validated on NASA SMAP and MSL (Curiosity) datasets.
- Built an explainability layer based on prediction-error attribution that ranks root-cause sensors without the compute cost of SHAP.
- Ingested live satellite frames through SatNOGS decoders and built a Streamlit/Plotly mission-control dashboard with severity scoring and root-cause views; containerized with Docker.

SentinelAI: Autonomous SIEM & SOAR Assistant | Python, LangGraph, Ollama, Redpanda, OpenSearch, Sigma, TheHive, Shuffle, Docker

- Built a containerized SIEM/SOAR pipeline that normalizes endpoint, network and cloud logs to the OCSF schema with Vector and Redpanda (Kafka) and indexes them in OpenSearch.
- Compiled Sigma rules into OpenSearch queries to detect MITRE ATT&CK techniques such as brute force and PowerShell download activity.

Dubai Multitemporal Urban Analysis (academic lab) | Google Earth Engine, Landsat 8/9

- Computed NDVI, NDBI and RGB composites from Landsat 8/9 imagery for Dubai (2015–2024) to study vegetation and built-up change.

CTF & HANDS-ON PRACTICE

Handle: ragefrost. Focus areas: web exploitation, SSRF chaining, JWT and prototype-pollution attacks, authentication bypass, Active Directory attacks.
Events: GenCysCTF 2.0, CloudSEK CTFx (Black Vault CTF), MythX National CTF (KIET).
Challenge author: Built "Merge Conflict" for crimsonCTF: a Node.js/Express chain of .git leak, JWT algorithm confusion, prototype pollution and EJS RCE.
TryHackMe: hands-on practice on guided paths and labs, tryhackme.com/p/Ragefrost

EXPERIENCE

Spatial Data Engineering Intern | Thazhal Geospatial June 2026

- Built a full-stack spatial AI tool (Next.js, Python, GeoPandas) that estimates Total Addressable Market for financial-infrastructure expansion.
- Wrote offline pipelines to ingest, clean and spatially aggregate 23,000+ commercial locations from the Google Places API into municipal boundaries.

CERTIFICATIONS

- Cybersecurity Bootcamp: Software Security and Vulnerability Assessment, IIITDM Kancheepuram
- AWS AI & ML Scholar and AWS AI Practitioner Challenge (Udacity, 2026)